

Energy-Efficient CSI-Based People Tracking with Spiking Neural Networks

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Abstract

Device-free indoor human monitoring often leverages Wi-Fi channel state information (CSI), extracted from commodity access points. However, the deep neural networks (DNNs) typically used for CSI-based sensing require a computational capacity exceeding what embedded and internet of things (IoT) devices can sustain. Spiking neural networks (SNNs) are a promising alternative, being naturally suited to temporal signals whose informative content lies in changes between consecutive channel measurements. In this work, we formulate people tracking as a joint presence-detection and position-estimation problem, and propose an SNN-based architecture that processes delta-encoded CSI amplitudes. The model is trained and evaluated on an IEEE 802.11ax CSI dataset spanning five activities in line of sight (LoS) and non-line of sight (NLoS) conditions. The ground truth is obtained through position estimation on a synchronized video. The pipeline: *i)* distinguishes occupied from empty rooms, and *ii)* estimates the tracked subject's approximate position from CSI alone. This work suggests SNNs are a low-power substitute for conventional networks in CSI-based sensing.

CCS Concepts

• **Computing methodologies** → **Bio-inspired approaches**; *Neural networks*; • **Networks** → **Network measurement**; Network privacy and anonymity; Wireless access networks.

Keywords

Channel state information, Spiking neural network, Wi-Fi, Joint communication and sensing, Indoor tracking

1 Introduction

Recent advances in joint communication and sensing (JCAS) have shown that commodity Wi-Fi infrastructure and hardware can integrate ambient sensing capabilities. In particular, channel state information (CSI) provides a fine-grained description of the wireless propagation channel, whose variations reflect changes in the surrounding environment. This enables applications ranging from human activity recognition (HAR) to occupancy detection, localization, and people tracking, using already-deployed communication hardware.

Most state-of-the-art CSI-based sensing tools rely on deep learning (DL) models to extract meaningful patterns from

the high-dimensional temporal structure of the channel. Albeit highly accurate, these approaches are computationally and energetically expensive, making them unsuitable for resource-constrained platforms where sensing applications are increasingly expected to operate. In these contexts, reducing resource requirements is as important as maximizing sensing accuracy.

Spiking neural networks (SNNs) offer a promising alternative: by communicating information through sparse events, they naturally reduce the number of operations performed during inference while preserving the ability to model temporal dynamics. In this framework, CSI-based sensing is particularly suited to event-driven processing, since sensing leverages *changes* in the wireless channel rather than its absolute value. However, the use of SNNs for CSI-based people tracking remains largely unexplored.

In this paper we investigate whether an SNN can provide an energy-efficient solution for device-free indoor tracking method. The main contributions of this work are:

- An SNN architecture for joint occupancy detection and continuous position estimation from Wi-Fi CSI;
- The validation of such architecture on a new 802.11ax CSI dataset consisting of ~ 200 minutes of channel monitoring in a real-world environment;
- The comparison of the proposed SNN with two comparable dense and recurrent neural networks (NNs), showing that it outperforms both while requiring orders of magnitude less energy.

2 Related Work

To frame this paper in the vast literature on JCAS, this section focuses on the two building blocks of this work: Wi-Fi CSI and SNNs.

2.1 CSI-based Indoor Sensing

Wi-Fi CSI is one of the most commonly used sources of information for JCAS. Its interpretation as an 'electromagnetic fingerprint' of the environment on the propagating signal offers an opportunity to exploit its information content for ambient sensing tasks. Among the most widespread applications, HAR and device-free localization and tracking are likely those that capture the most attention.

NN- or machine learning (ML)-based approaches are common in works aiming at environment-independent HAR. The authors of [7] extract micro-Doppler traces from CSI data

and feed them into a NN architecture to recognize dynamic activities, while [11] identifies a set of physically explainable features of the delay-Doppler domain representation of CSI to use for support vector machine (SVM)-powered HAR.

Indoor localization and people tracking studies, instead, rely more frequently on DL to extract distinctive information from the CSI. The authors of [13] devise a DL-based approach to differentiate the fingerprints associated to different user locations and reduce classification ambiguity; similarly, [14] proposes a CSI augmentation pipeline to enhance the distinguishability of the fingerprints used for localization. Models estimating the user velocity and location by integrating spatiotemporal Doppler information and angle of arrival of the received signal make indoor device-free user tracking possible [5]. Alternative approaches to this task involve leveraging the CSI difference for fine-grained Doppler speed estimation, resulting in a median tracking error as low as 34 cm [4].

For a broader overview of CSI-based JCAS applications, the interested reader can refer to recent works like [8].

2.2 SNN for JCAS

Training SNNs on real-valued data has become more practical with the development of spatio-temporal and surrogate gradient-based methods, which enable standard back-propagation [9]. SNNs have already been applied on CSI specifically, mostly using the computational constraints of Wi-Fi hardware as central motivation. For instance, [6] and [15] propose spiking convolutional architectures for Wi-Fi activity recognition using CSI. Moving beyond CSI, [3] applies a spiking network with unsupervised spike-timing-dependent plasticity to a related task, where Wi-Fi frames are detected from raw radio captures rather than channel state. CSI-based works typically frame their task as classification: a window of CSI samples is usually assigned just one activity label, not a continuous value. Continuous position estimation from CSI using SNNs remains largely unexplored, although the premises for it exist separately. One of the relevant examples is [1], where wearable sensor data are encoded by change rather than absolute value, and SNN energy is measured directly on neuromorphic hardware. The authors of [10] decode the continuous position and velocity from spiking activity, matching long short-term memory (LSTM) precision. Whether these ideas combine on CSI specifically is still an open question.

3 Principles of Wi-Fi CSI-based sensing

The CSI is computed at the receiver from a known frame preamble as the ratio between the received symbols and the expected ones, providing an approximation of the channel frequency response. For orthogonal frequency division multiplexing (OFDM)-based technologies with N_k subcarriers and

N_{SS} spatial streams, one CSI is extracted per spatial stream. The CSI is represented as an array of N_k complex numbers, with the n -th element corresponding to the IQ component of the channel response associated to the n -th subcarrier. Some subcarriers are suppressed in transmission and are instead used as guard bands or for pilot symbols, therefore the CSI value is set to $0 + i0$ on the corresponding positions. The IQ complex values are commonly converted into the equivalent representation using amplitude and phase and undergo a normalization or cleaning process; the use of either or both components for sensing is application-dependent.

The CSI cleaning techniques used in this work aim at reconstructing both phase and amplitude on the suppressed subcarriers while ensuring a low computational complexity for the preprocessing step. In the IEEE 802.11ax standard, an 80 MHz channel consists of 1024 subcarriers: the external guard bands involve subcarriers $[-512, \dots, -501]$ and $[+502, \dots, +511]$, while the central null interval takes up subcarriers $[-2, \dots, +2]$. The central nulls generate a discontinuity on the frequency axis, thus their ‘expected’ amplitude and phase values need to be reconstructed by interpolation of the adjacent data subcarriers. On the contrary, the zero values in the external guard bands need no patching, since they do not create gaps in the frequency domain. Amplitude is also affected by the automatic gain control (AGC), which is compensated for by normalizing the energy of each CSI; finally, phase is aligned at zero on the central subcarrier.

4 Dataset Experimental Setting

The dataset considered in this study contains CSI samples collected across multiple days in the month of June 2025. All experiments are thus subject to real-world variability due to minor modifications of the experimental environment from one day to the other; for instance, small pieces of furniture may be displaced during the interval between two measurement sessions. This property of the dataset is intentional and highlights the adaptability of the proposed sensing methodology to a variable and realistic input.

The selected indoor environment is an everyday office (Fig. 1), wherein the transmitter (Tx) and receiver (Rx) are placed in fixed positions at two opposite ends of the room. The Tx is placed on a 40 cm-high support behind a drywall partition, creating partial line of sight (PLOS) signal propagation conditions. The Rx is placed on a table beside a mobile phone (‘P’ in Fig. 1) used to film the scene during the measurement sessions, providing a ground truth reference. When non-line of sight (NLOS) propagation conditions are tested, a metallic shield (‘S’ in Fig. 1) is placed in front of the Tx.

CSI samples are collected using Nexmon CSI Extractor tool, which allows to fully customize the experimental configuration of 802.11ax frames [2]. Tx and Rx communicate on the

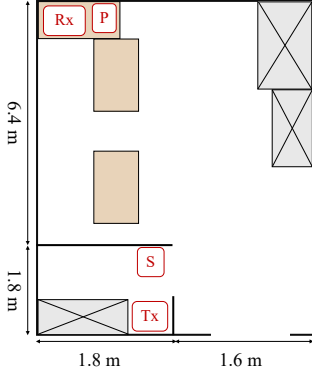


Figure 1: Floor plan of the experimental environment.

80 MHz 802.11ax channel with center frequency 5.775 GHz. Both devices are commercial off-the-shelf (COTS) Asus access points (APs): the Tx is an RT-AX82U, while the Rx is an RT-AX86U; both are equipped with Wi-Fi Broadcom chipset BCM43684. One antenna is active on the Tx, while three receiving antennas are enabled on the Rx. All devices are synchronized via network time protocol (NTP).

Hardware limitations impose a maximum sampling rate of 200 frames per second to ensure lossless CSI collection. To show the resilience of the proposed sensing tool across variable frame rates, as is common in real traffic conditions, the dataset consists of CSI collections obtained with different sampling rates: artificial traffic is transmitted on the channel, with frames interspaced by $\Delta t = 5, 10, 50, 100$ ms. Specifically, $\Delta t = 100$ ms imposes the same frame rate as standard Wi-Fi beacons, which helps highlighting the feasibility of the proposed approach using a default traffic rate in Wi-Fi.

For each Δt value five sets of experiments are performed, encompassing different combinations of activities: empty room (E), a person entering the empty room and sitting at the desk closer to the Rx (EN-S), a person entering the empty room and walking along a random trajectory (EN-W), a person sitting at the desk, then standing up and leaving the room (L), a person sitting at the desk (S).

5 The SNN Model

5.1 Problem Formulation

We cast people tracking as a joint presence-detection and position-regression problem. Given a window of T feature frames $F[1], \dots, F[T]$ (Section 5.3), the SNN outputs at every timestep t are a presence estimate $\hat{p}[t]$ and a position estimate $\hat{y}[t]$, instead of a single label per window. The network jointly minimizes a presence classification loss and a position regression loss,

$$\min_w \mathbb{E} [\mathcal{L}_{\text{presence}} + \mathcal{L}_{\text{position}}], \quad (1)$$

where $\mathcal{L}_{\text{presence}}$ penalizes disagreement between $\hat{p}[t]$ and the ground-truth presence label of Section 5.2, and $\mathcal{L}_{\text{position}}$ penalizes the distance between $\hat{y}[t]$ and the ground-truth position, evaluated only over frames where the subject is present. Training details are given in Section 5.4.

5.2 Ground-Truth Labels

Presence and position labels come from the synchronized room-camera video (Section 4), processed with MediaPipe’s pose landmarker. Shoulders and ears are tracked because they stay visible also when the subject faces away. The position label $y[t]$ is the head/shoulder midpoint in normalized image-plane coordinates, $(x, y) \in [0, 1]^2$ and no camera calibration or room geometry is used to convert this into a metric position.

The landmarker can fail to detect a pose, but labeling every such frame as “absent” would be wrong. Detection gaps at the beginning or the end of a trial reflect real absence, since the person has either not yet entered or has already left; gaps surrounded by detections are treated as temporary tracking failures and labeled present. This distinction is the most impactful for the L and S activities, where the person sits still for long periods and the detector is likely to lose its track.

Short position gaps are filled with linear interpolation between the neighboring detections, while long gaps are left undefined and excluded, regardless of the presence label. The resulting trace is smoothed with a short moving average and resampled to the CSI frame rate, giving the per-timestep presence label and position target.

5.3 Feature Engineering

Let $x_{a,k}[t] \in \mathbb{C}$ denote the raw CSI value on antenna $a \in \{1, \dots, N_a\}$, subcarrier $k \in \{1, \dots, N_k\}$, at frame t ($N_a = 3$, $N_k = 1024$). All features are based on two primitives: per-subcarrier amplitude $|x_{a,k}[t]|$, and the phase of $x_{a,k}[t]$ relative to a reference antenna.

Amplitude is normalized per frame to unit energy across subcarriers,

$$\tilde{a}_{a,k}[t] = \frac{|x_{a,k}[t]|}{\sqrt{\sum_{k'} |x_{a,k'}[t]|^2}}, \quad (2)$$

to make it is robust to AGC variations across sessions.

All antennas share one local oscillator, so that it carries a common carrier/sampling-offset drift, which is not connected to motion. This drift cancels in the conjugate product between antenna a and a fixed reference antenna (antenna 1),

$$u_a[t, k] = \frac{x_{a,k}[t] x_{1,k}^*[t]}{|x_{a,k}[t] x_{1,k}^*[t]|}, \quad a \neq 1, \quad (3)$$

retained as $(\text{Re } u_a, \text{Im } u_a) = (\cos \Delta\phi_a, \sin \Delta\phi_a)$. Otherwise, discontinuities due to phase wraps would be misread as

motion. With $N_a = 3$, this gives 3 amplitude channels and $(N_a - 1) \times 2 = 4$ phase channels: 7 channels per subcarrier per frame and the three motion-specific channels below.

Motion magnitude. Presence and position depend on the channel's *change* rather than its absolute state. We refer to this change as *motion magnitude*, though it is a property of the CSI signal, not a direct measurement of the subject's physical motion. The average frame-to-frame amplitude change across antennas and subcarriers,

$$m[t] = \frac{1}{N_a N_k} \sum_{a,k} ||x_{a,k}[t]| - |x_{a,k}[t-1]|, \quad (4)$$

is broadcast as a single additional channel across all subcarriers.

We also consider a normalized variant, dividing $m[t]$ by the median of m over a trailing 20-frame window (≈ 1 s),

$$\tilde{m}[t] = \frac{m[t]}{\text{median}(m[t-w+1], \dots, m[t]) + \epsilon}, \quad (5)$$

since $m[t]$ inherits amplitude's sensitivity to AGC gain drift. Normalizing against a local median keeps $\tilde{m}[t]$ well-behaved as gain drifts within a trial, without being distorted by isolated motion spikes inside the window itself.

Spectral motion ratio. Motion magnitude alone cannot separate genuine motion from a noisier sensor, since both increase $m[t]$. The two differ in *how* the signal varies over time, not *how much*: human motion is rhythmic and concentrated at low frequencies (footsteps at 1–2 Hz; sway and breathing even lower), while sensor and AGC noise is approximately white, spread evenly across frequencies regardless of its overall level. We exploit this by comparing energy in a low, motion-associated frequency band against a high, noise-reference band, computed per frame from the power spectrum of the antenna- and subcarrier-averaged amplitude over a trailing, Hann-windowed 3.2 s segment (at the pipeline's 20 Hz frame rate),

$$r[t] = \log_{10} \left(\frac{P_{[0.15,3] \text{ Hz}}[t]}{P_{[5,10] \text{ Hz}}[t]} + \epsilon \right). \quad (6)$$

Unlike $m[t]$, a rise in the noise floor raises both bands together and leaves $r[t]$ largely unaffected; the logarithm keeps this otherwise wide-ranging ratio on a scale comparable to the pipeline's other, roughly unit-scale channels. The window resolves footstep-band motion but not slower breathing, which would require a substantially longer window.

Cross-antenna coherence. The spectral ratio characterizes a single averaged signal. A complementary criterion is agreement between independent antennas: a body in the room reflects the same physical wavefront to all three antennas simultaneously, so genuine motion produces correlated fluctuations across antennas at shared frequencies, whereas

each antenna's thermal and hardware noise is independent. We measure this with the magnitude-squared coherence between antennas. Let $S_a[t, f]$ denote the short-time spectrum of antenna a 's amplitude trace at frequency f , computed causally over a trailing 6.4 s window. A coherence estimate from a single spectral snapshot is trivially 1 by construction, so S_a is instead computed over several overlapping sub-segments of that window (Welch's method), and coherence between antennas a, b is estimated as

$$\gamma_{ab}^2[t] = \left\langle \frac{|\langle S_a S_b^* \rangle|^2}{\langle |S_a|^2 \rangle \langle |S_b|^2 \rangle} \right\rangle_{\text{band}}, \quad (7)$$

where $\langle \cdot \rangle$ averages over sub-segments and $\langle \cdot \rangle_{\text{band}}$ over the motion frequency band. $\gamma_{ab}^2[t]$ is averaged over all antenna pairs to give one coherence value per frame. High coherence in the motion band indicates a shared physical event; low coherence indicates uncorrelated per-antenna noise, regardless of its magnitude.

Motion magnitude, spectral ratio, and coherence are each a single scalar per frame, broadcast uniformly across all 1024 subcarriers.

5.4 Network Architecture

The feature stack $F[t]$ of Section 5.3 is segmented into sliding windows of $T = 64$ frames (stride 32, 50% overlap – 3.2 s clips at the 50 ms capture rate) and passed through a per-frame 1D convolutional encoder, two layers of 16 and 32 channels, kernel size 9, each followed by rectified linear unit (ReLU) and $2 \times$ max-pooling, applied independently to every frame's (channel $\times$ subcarrier) slice.

The resulting compact per-frame feature vector $f[t]$ is converted into a binary spike train through threshold-crossing (delta) encoding,

$$s_i[t] = \begin{cases} 1, & f_i[t] - f_i[t-1] \geq \theta \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

with $\theta = 0.3$, preferred over Poisson rate coding since the motion information CSI carries lies in how the channel changes over time and not in its instantaneous magnitude.

The spike train $s[t]$ feeds a two-layer leaky integrate-and-fire (LIF) trunk [?], shared between both output heads, with membrane decay $\alpha = 0.9$ and firing threshold $\vartheta = 1.0$, trained end-to-end via a fast-sigmoid surrogate gradient. The two LIF layers have size 128 and 32, respectively.

Presence and position are read out directly from the second LIF layer's *spike* output $a_{2,t}$, densely at every one of the window's T timesteps rather than once per window, through two small heads,

$$\hat{p}[t] = W_p a_{2,t} + b_p, \quad (9)$$

$$\hat{y}[t] = \text{MLP}_y(a_{2,t}), \quad (10)$$

where $\hat{p}[t]$ is a presence logit, $\hat{y}[t] \in \mathbb{R}^2$ an (x, y) position estimate, and MLP_y a single hidden layer of width 16 with a ReLU nonlinearity (multi-layer perceptron (MLP)). Both heads operate on the same binary, 32-dimensional code at each 50 ms frame, rather than a membrane potential accumulated over the whole window.

The network is trained end-to-end on the objective of Eq. (1) with Adam (learning rate 5×10^{-4} , batch size 32) for a fixed 60 epochs per fold; the checkpoint with the lowest validation loss is retained for testing, with no separate early-stopping rule beyond this post-hoc selection.

Model performance is evaluated via leave-activity-out cross-validation. Of the five activity codes, only the four with a person present (EN-S, EN-W, L, S) are ever held out; the fifth, the empty-room code E, stays in every fold’s training set so the model always sees genuine absent-class examples. Each of the four folds holds out one activity’s two sessions entirely as the test set, uses a *different* activity’s two sessions for validation (checkpoint selection and presence-threshold calibration), and trains on the remaining six sessions; training sees only three of the five activity codes per fold, while one further activity is set aside for validation and is never trained on either.

6 Results

We compare SNN model (amplitude+phase+cross-coherence, Section 5.3) against two non-spiking benchmarks trained on the identical feature set and evaluated under the same leave-activity-out cross-validation of Section 5.4 (4 folds): an artificial neural network (ANN) and an LSTM, both built on the same convolutional encoder and trunk sizes as the SNN. Table 2 reports presence accuracy, position root mean squared error (RMSE), and per-window energy, as mean $\pm$ standard deviation across folds.

RMSE is $\sqrt{\mathbb{E}[\|\hat{y}[t] - y[t]\|_2^2]}$, averaged over present-labeled frames. Since $y[t]$ is in normalized image-plane coordinates (Section 5.2), RMSE is a unitless average positional error and not a metric distance, meaning that converting to real units would need camera calibration this pipeline does not have.

Energy follows the standard multiply-accumulate/accumulate-only model (45 nm CMOS): 4.6 pJ per dense operation (ANN/LSTM) versus 0.9 pJ per spike-gated operation (SNN), charged only on a spike – 0.2%–4% per-layer spike rates account for most of the gap. The convolutional front-end is excluded from all counts; this is a per-inference operation-count estimate, not a cycle-accurate simulation.

The SNN gives the best presence accuracy and RMSE of the three, at $2.4 \times 10^3 \times$ less energy per window than the ANN twin and $9.6 \times 10^3 \times$ less than the LSTM (Table 2). The explanation of the energy gap doesn’t lie in the model size: SNN

Table 1: Comparison of the best model against two benchmarks.

Model	Presence Acc.	RMSE	Energy/Win.
SNN	0.753 ± 0.081	0.2363 ± 0.0849	1.3×10^2 nJ
ANN twin	0.472 ± 0.283	0.3607 ± 0.0584	3.1×10^5 nJ
LSTM	0.587 ± 0.195	0.2889 ± 0.1153	1.3×10^6 nJ

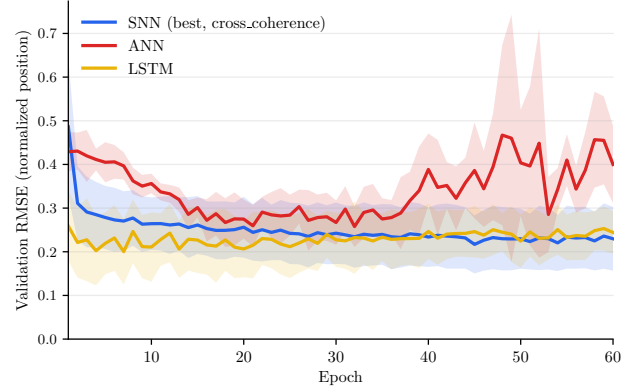


Figure 2: Validation RMSE across training epochs for the SNN, ANN, and LSTM.

Table 2: Feature ablation: each row adds one feature to amplitude+phase (Eqs. (2) and (3)). Leave-activity-out CV, mean $\pm$ std, 4 folds.

Added Feature	Presence Acc.	RMSE
amp+phase only	0.648 ± 0.073	0.3669 ± 0.1439
+ motion magnitude (Equation (4))	0.646 ± 0.050	0.2606 ± 0.0793
+ spectral ratio	0.690 ± 0.054	0.2455 ± 0.0843
+ cross-coherence	0.753 ± 0.081	0.2363 ± 0.0849

and ANN share an identical architecture (1 059 235 parameters each), differing only in spiking (LIF) vs. dense (ReLU) activation, so the gap isolates activation sparsity as the cause. LSTM’s larger parameter from its four gated weight matrices independently explains its higher energy cost. The ANN’s high accuracy variance (0.472 ± 0.283) indicates less stable fold-to-fold behavior than either recurrent model, consistent with having no reset mechanism after high activation.

6.1 Feature Ablation

Table 3 isolates the contribution of each engineered feature of Section 5.3, added individually on top of amplitude+phase.

RMSE improves monotonically with each addition; accuracy dips slightly when raw motion magnitude is added, then recovers and improves further with spectral ratio and cross-coherence, which gives the largest improvement and motivates its choice as the primary configuration.

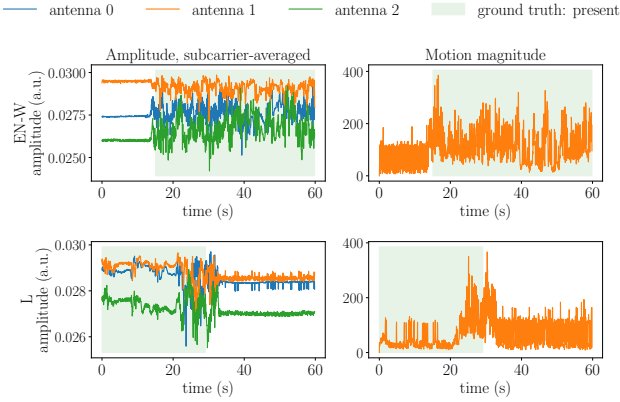


Figure 3: Amplitude and motion magnitude ($m[t]$, Equation (4)), PLoS, for EN-W (top; continuous motion) and L (bottom; motion drops to near-zero while seated, despite continued presence).

Raw motion magnitude $m[t]$ and spectral ratio are limited by activity. L (sits, then stands and leaves) is the clearest case, since during the sitting, motion-based features go quiet even though ground truth still labels the subject present, which is the same presence-detection problem handled by the gap-aware relabeling of Section 5.2. Figure 2 contrasts this against EN-W, where motion is continuous.

7 Conclusion

This paper presented an energy-efficient approach to device-free indoor people tracking through Wi-Fi CSI using SNNs. We formulated the task as a joint presence-detection and position-regression problem and proposed a feature representation combining normalized CSI amplitude, inter-antenna phase differences, and motion-aware features. The evaluation on an IEEE 802.11ax dataset collected under both PLoS and NLoS conditions showed that the proposed SNN outperforms comparable dense and recurrent NNs while reducing inference energy by several orders of magnitude. These results suggest that SNNs are a promising alternative to conventional DL for resource-constrained platforms.

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